SchoolRoute TN — Visual Project Walkthrough

Smart Digital Transit & Attendance Safety Monitoring System for Government School Commuters

G.O. Ms. No. 42 COMPLIANT | ZERO-PHONE ARCHITECTURE | MoRTH AIS-140 PANIC HARDWARE | LIVE CELLULAR GSM SMS DISPATCH

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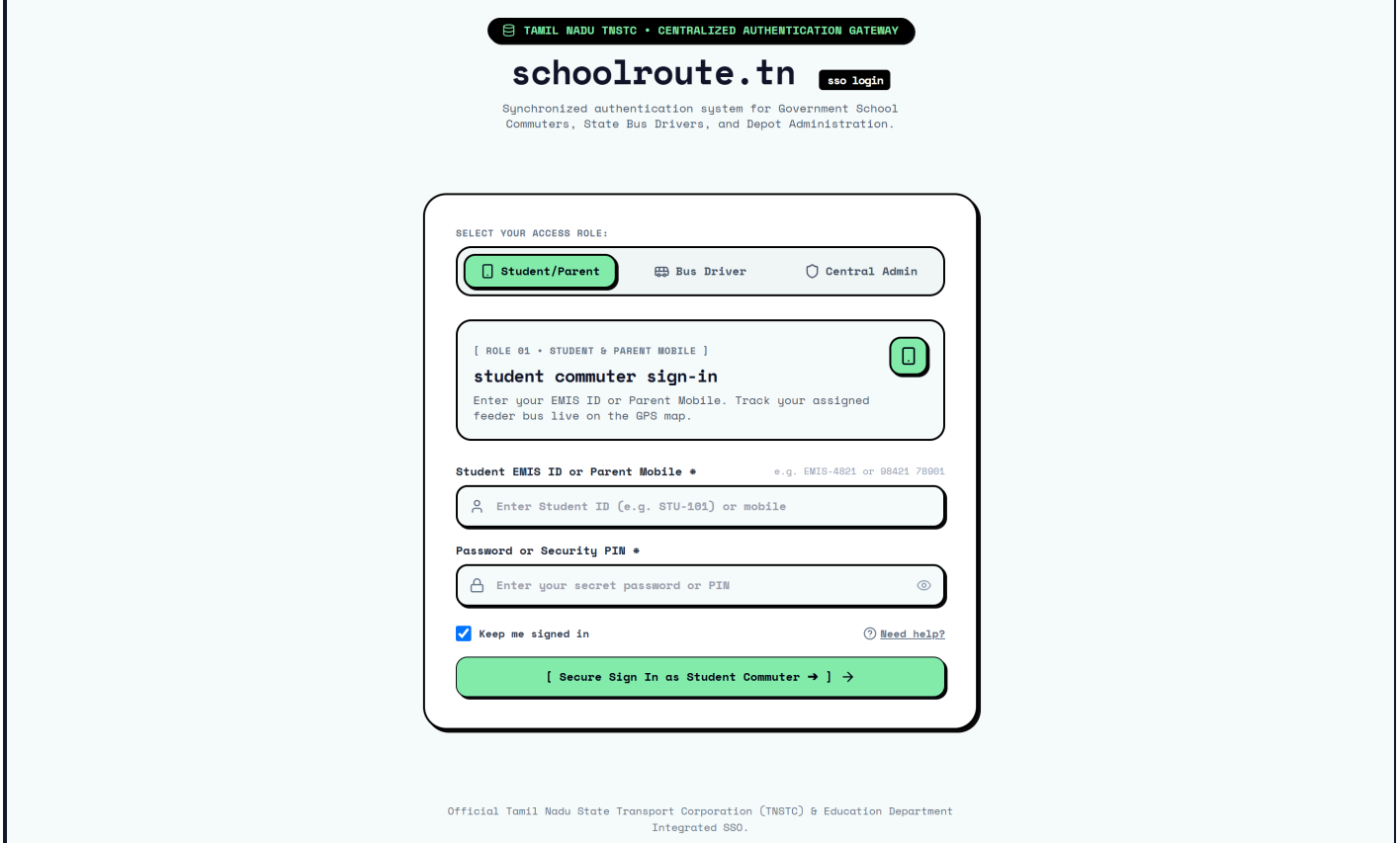
Executive Summary: This document demonstrates the complete execution and running interface of **SchoolRoute TN**. Designed specifically for underprivileged Tamil Nadu government school students traveling under the 100% Free Commuter Bus Scheme, it completely eliminates the need for students to carry smartphones. Using printable laminated 2D QR bus passes, camera-based driver verification (jsQR), on-bus AIS-140 physical panic buttons, and real-time cellular SMS dispatch via Vercel serverless gateways, the platform ensures total commuter safety and attendance integrity.

Role / Portal	Primary User	Hardware Utilized	Core Operational Capability
Student & Parent Mobile Portal	School Commuters & Parents	Physical Laminated QR Pass & Basic Keypad/GSM Phone	Real-time Leaflet GPS bus tracking, digital pass generator, AIS-140 on-bus emergency panic simulator, live SMS log.
Driver Cabin Console	TNSTC State Bus Driver	Standard Mobile/Tablet with Camera	Continuous jsQR camera pass scanner, trip controls, live GPS speed/crowd telemetry, cabin flashing emergency siren.
Admin & HQ Command Desk	District Transport Officer (DTO)	Depot Computer / Browser Console	Statewide fleet radar, student roster management, dynamic driver duty allocation, grievance redressal & official petition generator.

SCREEN 1: CENTRALIZED ROLE-BASED AUTHENTICATION GATEWAY

The entry point to the system is a high-contrast Neo-Brutalist Single Sign-On (SSO) gateway allowing instant, secure role-based access across all three user classes.

SCREENSHOT 01 • ROLE LOGIN PORTAL URL: http://localhost:5173/



1. Role-Specific Tab Selector

Clean toggle switches between: (1) Student Commuter / Parent, (2) Bus Driver / Crew, and (3) Depot Transport Officer (Admin).

2. Commuter Identification Verification

Students authenticate via their unique EMIS ID (e.g. `STU-101`) or parent registered mobile phone (+91 73059 40480) with student PIN.

3. Pre-defined Driver Fleet Verification

Crew login verifies driver registration against the TNSTC Depot roster, automatically synchronizing the driver's active route schedule.

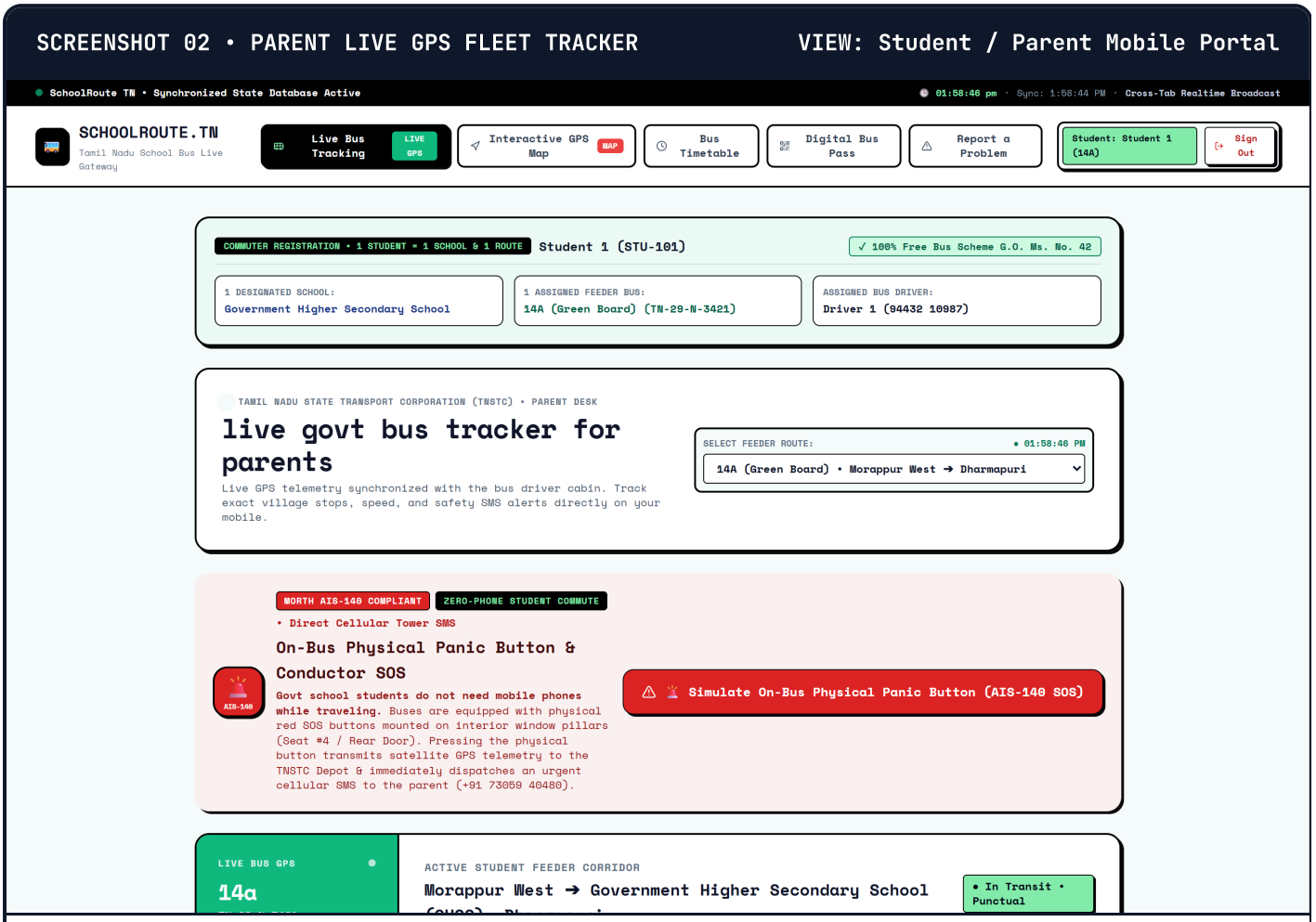
4. District Transport Officer Security

Central administrative master login reserved for the District Transport Officer with statewide oversight credentials.

Engineering Details: Implemented in `src/pages/RoleLoginPage.jsx` with session persistence stored in `localStorage` under `schoolroute_tn_active_role_clean_v7`.

SCREEN 2: PARENT LIVE GPS SATELLITE FLEET TRACKER

Real-time parent monitoring console showing feeder bus progress, distance countdown, telemetry statistics, and the live commuter boarding roster.



1. Live Progress & Remaining Distance

Calculates exact kilometers remaining to the school gate with live estimated time of arrival (ETA) updated as the bus moves.

2. Route Telemetry KPI Cards

Live displays for safe operating speed (e.g., 38 km/h), passenger crowd status, and assigned TNSTC depot details.

3. Stop-by-Stop Visual Progress Bar

Dynamic timeline showing completed stops (Morappur, Kadayampatti), active next stop, and upcoming destination school.

4. Commuter Boarding Status Roster

Displays whether the enrolled student is currently *Waiting at Stop* or verified *On Board*.

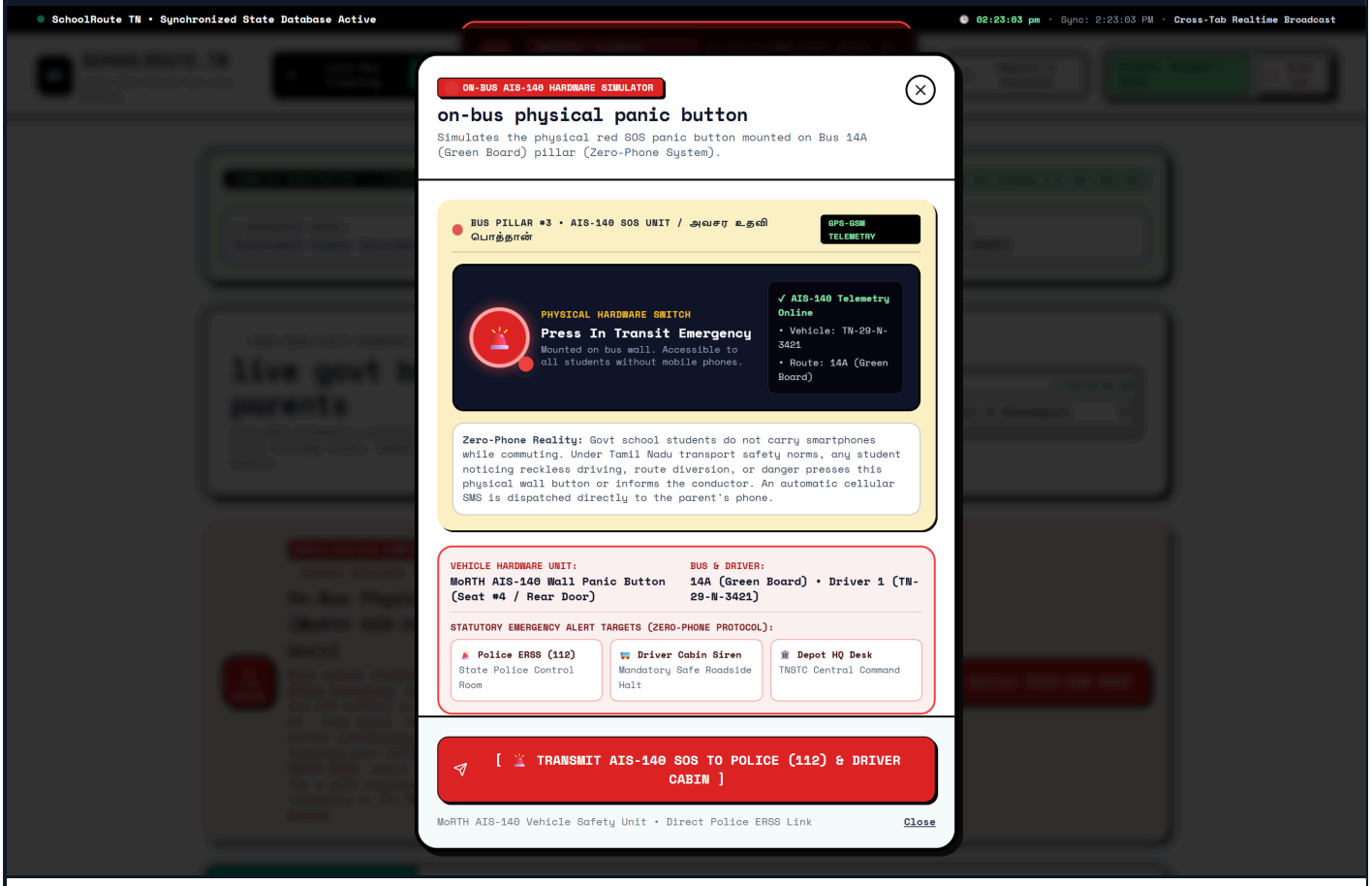
Engineering Details: Implemented in `src/components/ParentLiveTracker.jsx`, utilizing `LiveDatabaseContext` for reactive real-time updates.

SCREEN 3: MORTH AIS-140 ON-BUS PHYSICAL PANIC BUTTON SOS SIMULATOR

Physical vehicle emergency telemetry simulation that protects students traveling without mobile phones in cases of dangerous driving, harassment, or route deviations.

SCREENSHOT 03 • AIS-140 EMERGENCY SOS CONSOLE

MODAL: On-Bus Stanchion Panic Hardware



1. Wall-Mounted Physical Button Simulation

Simulates tactile red SOS push-buttons mounted on interior stanchion pillars adjacent to Seat #4 and the rear emergency door.

2. Four Emergency Incident Categories

Allows immediate categorization: (1) Rash Driving & Overspeeding, (2) Route Deviation, (3) Vehicle Breakdown, (4) In-Transit Harassment.

3. Statutory MoRTH AIS-140 Targets

Because the wall button is an anonymous vehicle hardware device, it does not identify individual passengers. Instead, it transmits direct satellite telemetry to: (A)
Police Control Room (Dial 112 / ERSS)
, (B)
Driver Cabin Flashing Siren
(prompting an immediate safe roadside pull-over), and (C)
TNSTC Feeder Depot Command Center
. (Individual parent SMS alerts are reserved for personal QR pass boarding and arrival events).

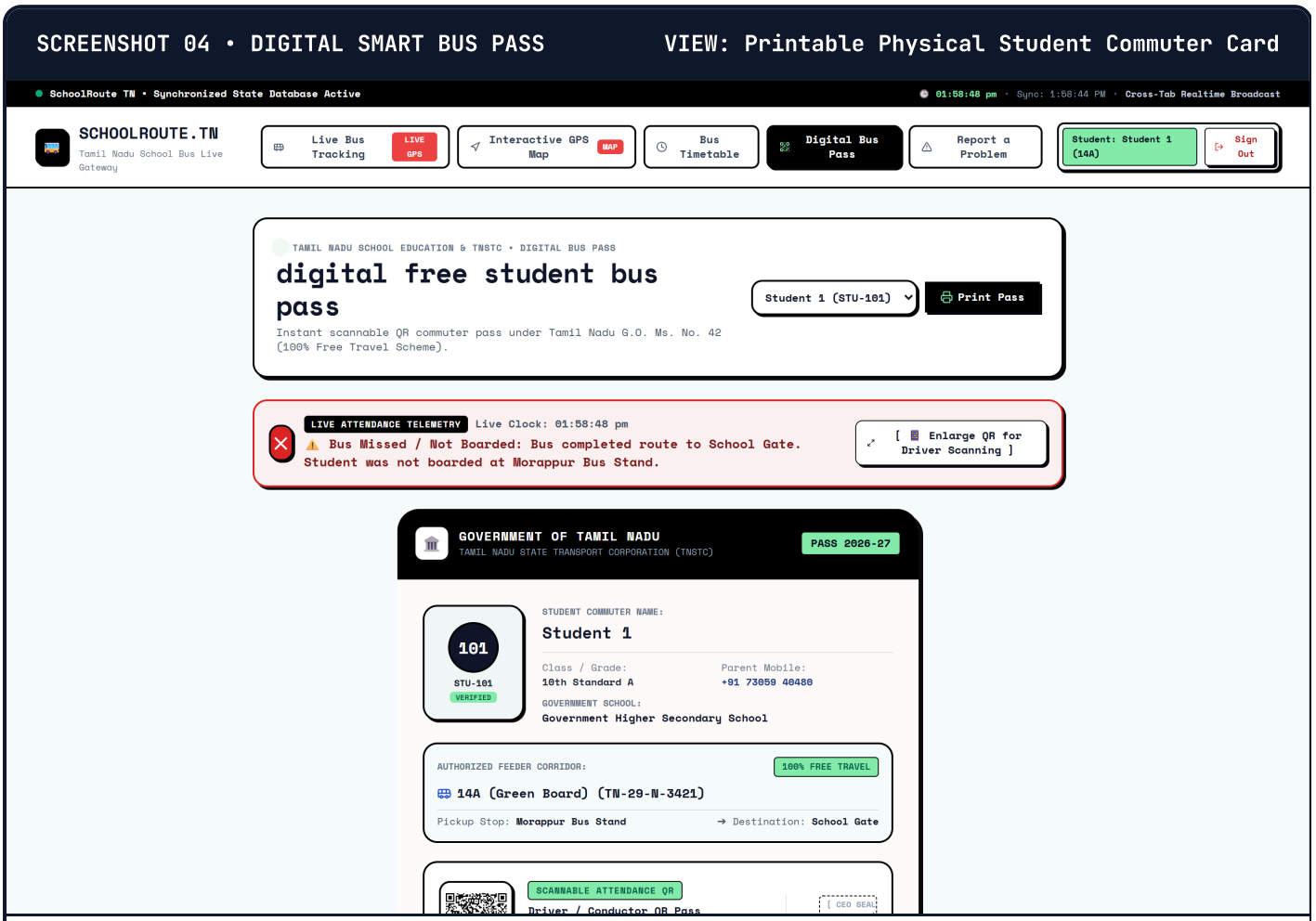
4. Statutory Helpline Integration

Direct institutional emergency numbers: Police Emergency 112, Childline 1098, Women Safety Helpline 181, and District Transport Control (04342-260100).

Engineering Details: Powered by `studentReportIrregularBehavior()` in `LiveDatabaseContext.jsx`, routing telemetry to Police ERSS 112 and driver dashboard.

SCREEN 4: STATUTORY DIGITAL STUDENT BUS PASS (G.O. MS. NO. 42)

Statutory Tamil Nadu Government School Free Bus Pass generator embedding tamper-resistant 2D QR matrix tokens for physical lamination.



1. Reed-Solomon 2D QR Token

Generates standardized QR tokens containing encrypted student data (EMIS ID, student name, assigned bus, pickup stop, parent phone).

2. Zero-Phone Commute Physical Pass

Designed to be printed on standard paper/cardstock and laminated; completely zero battery and zero internet requirement for the child.

3. Statutory Scheme Compliance

Officially formatted according to Tamil Nadu Government Order G.O. Ms. No. 42 (100% Free Concessional Travel Scheme).

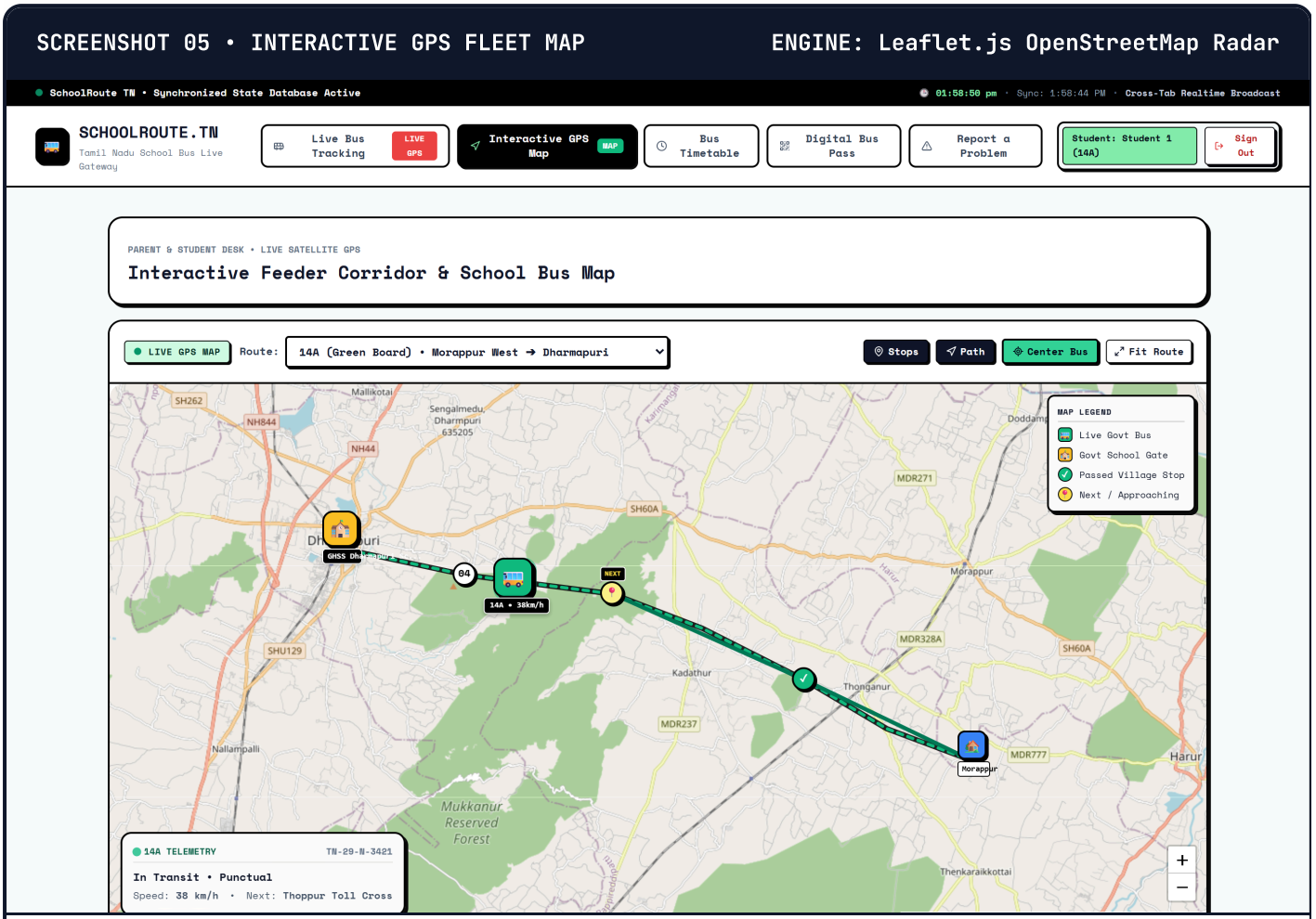
4. Instant Live Status Reflection

Live card badge displays real-time trip state: *Waiting at Stop*, *On Board*, or *Safely Arrived at School Gate*.

Engineering Details: Implemented in `src/components/StudentBusPass.jsx` using the `qrcode` NPM package with Error Correction Level M.

SCREEN 5: INTERACTIVE GIS FEEDER ROUTE MAP

Interactive OpenStreetMap radar powered by Leaflet.js rendering live vehicle positions, route polylines, feeder village halts, and destination schools.



1. Dynamic Bus Vehicle Marker

Pulsing vehicle icon positioned at live GPS coordinates along the state highway corridor with route direction orientation.

2. Feeder Stop Geocoding

Color-coded stop markers indicating passed stops (green), current stop, and destination government higher secondary schools.

3. Smooth Route Polyline

High-contrast path overlay mapping the exact legal route traveled by TNSTC school feeder buses.

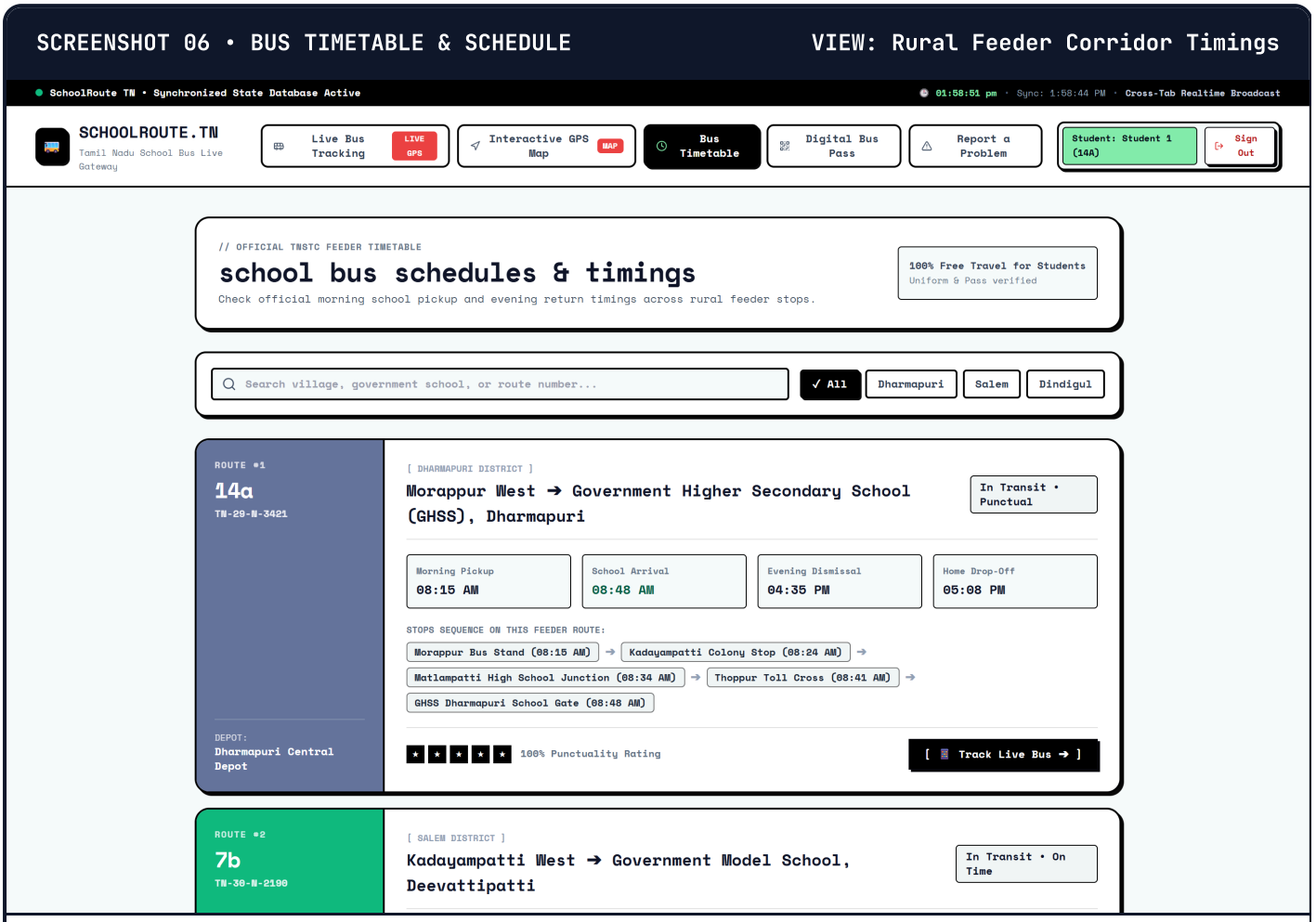
4. Interactive Popup Details

Clicking markers reveals pickup timings, distance milestones, student count, and driver emergency contacts.

Engineering Details: Implemented in `src/components/LiveBusMap.jsx` using standard OpenStreetMap tiles, custom Leaflet SVG div-icons, and dynamic bounds fitting.

SCREEN 6: FEEDER BUS TIMETABLE & CORRIDOR SCHEDULE

Public commuter schedule explorer detailing morning school pickup runs, evening return trips, intermediate village halts, and scheduled driver assignments.



1. Dual-Shift Transit Timing

Displays Morning Inbound School Feeder runs (07:30 AM – 08:30 AM) and Evening Outbound Return runs (04:15 PM – 05:15 PM).

2. Punctuality & Sync Indicators

Highlights synchronized connections preventing students from missing morning prayer bells or examination sessions.

3. Multi-Route Coverage

Enables switching between Route 14A (Morappur – Dharmapuri), Route 7B (Kadayampatti), and Route 21 (Pennagaram).

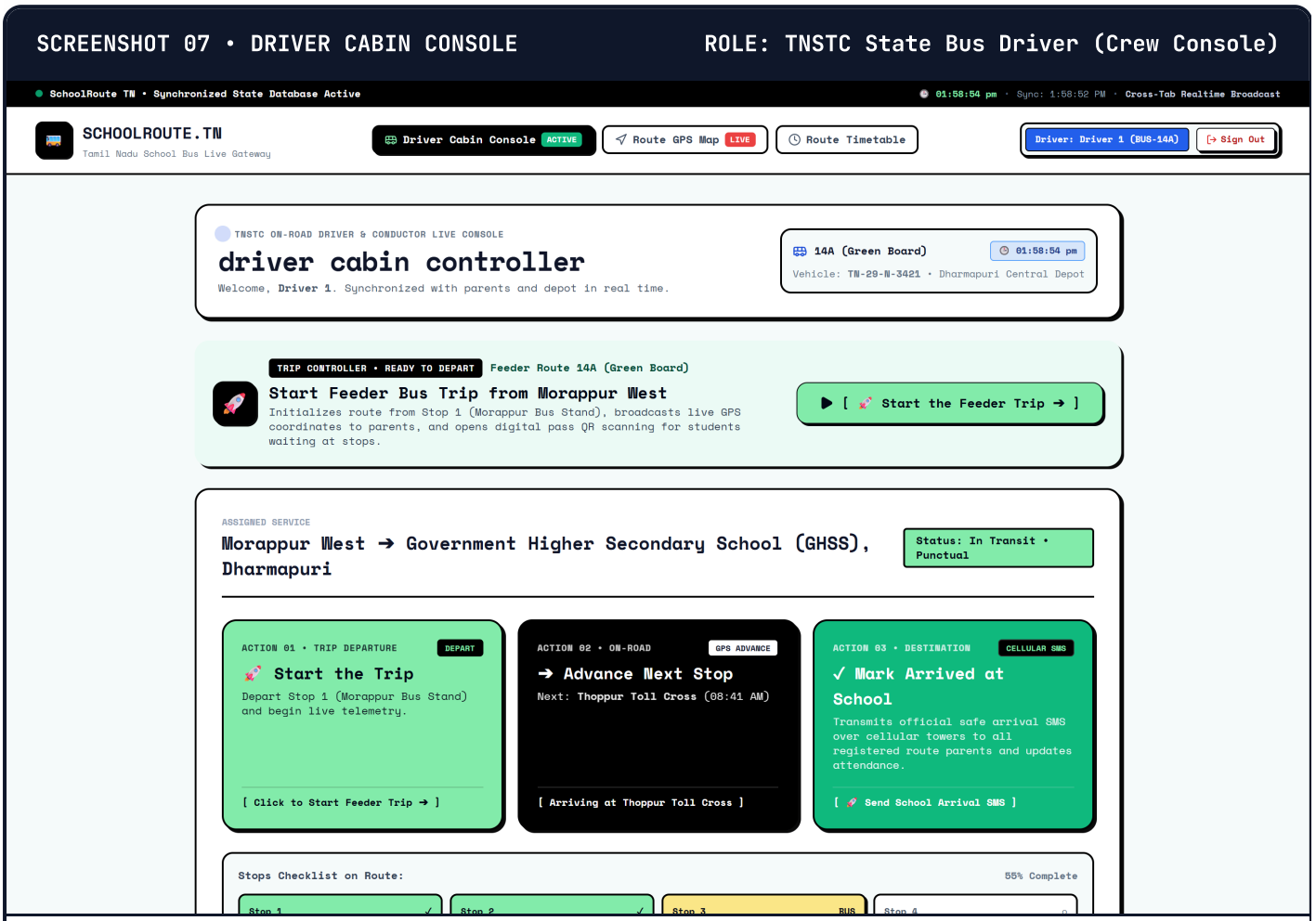
4. 1-Click Route Tracking

Quick action button to instantly track any timetable route directly on the live satellite map.

Engineering Details: Implemented in `src/components/BusTimetable.jsx` with static and live schedule synchronization.

SCREEN 7: BUS DRIVER CABIN TELEMETRY & ATTENDANCE CONSOLE

Dedicated mobile/tablet workspace engineered for state bus drivers with large tactile controls, live trip telemetry, delay reporting, and passenger boarding logs.



1. One-Click Trip Controls

Large touch buttons to "Start Feeder Trip", "Advance Next Stop", and "Mark Bus Arrived at School Gate".

2. Speed & Crowd Telemetry Sliders

Enables driver to update operating speed (km/h) and crowd level (Normal / Packed / Overcrowded) with instant cloud broadcast.

3. Real-Time Delay Broadcasting

Pre-configured delay triggers (Railway Gate Closed, Heavy Highway Congestion, Tire Repair) updating parent map indicators.

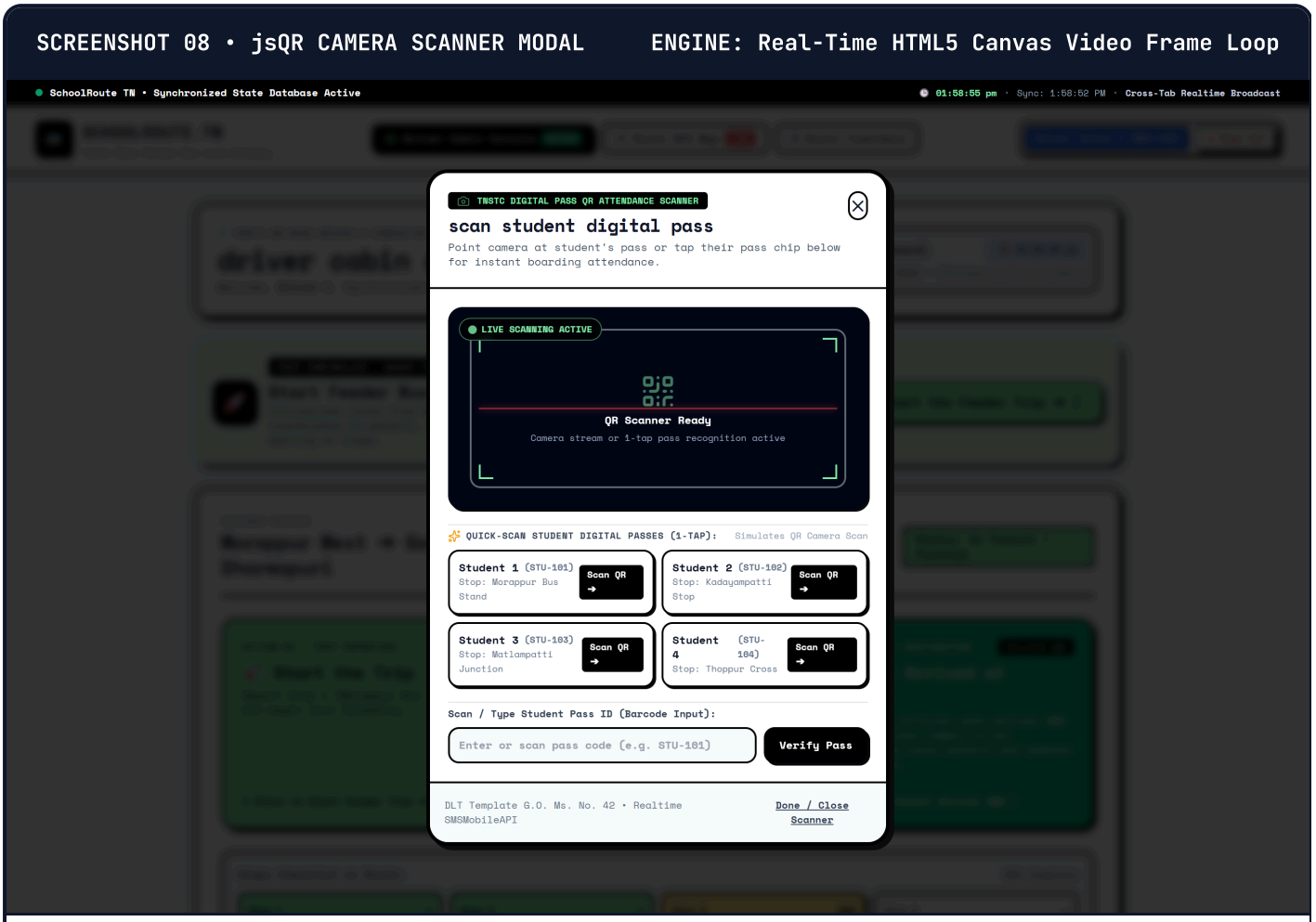
4. Commuter Boarding Roster

Interactive student roster with one-click manual boarding toggle for students whose physical cards are temporarily lost.

Engineering Details: Implemented in `src/components/DriverConsole.jsx`; state synced across all sessions via `LiveDatabaseContext.jsx`.

SCREEN 8: IN-BROWSER JSQR VIDEO STREAM CAMERA SCANNER

High-speed HTML5 Canvas camera scanner using the jsQR library to scan student passes on the bus entry step within milliseconds without freezing.



1. Continuous Non-Blocking Video Loop

Runs `requestAnimationFrame(scanVideoLoop)` unconditionally on every frame; unrecognized or damaged codes never freeze the stream.

2. Sub-Second Boarding Verification

Instantly extracts the encrypted payload, marks student as *On Board*, and dispatches a cellular SMS to parent mobile.

3. Frame Throttling & 2-Second Debounce

Prevents rapid duplicate scans and multiple SMS charges using timestamp cooldowns and React `useRef` locks.

4. Manual EMIS ID Fallback

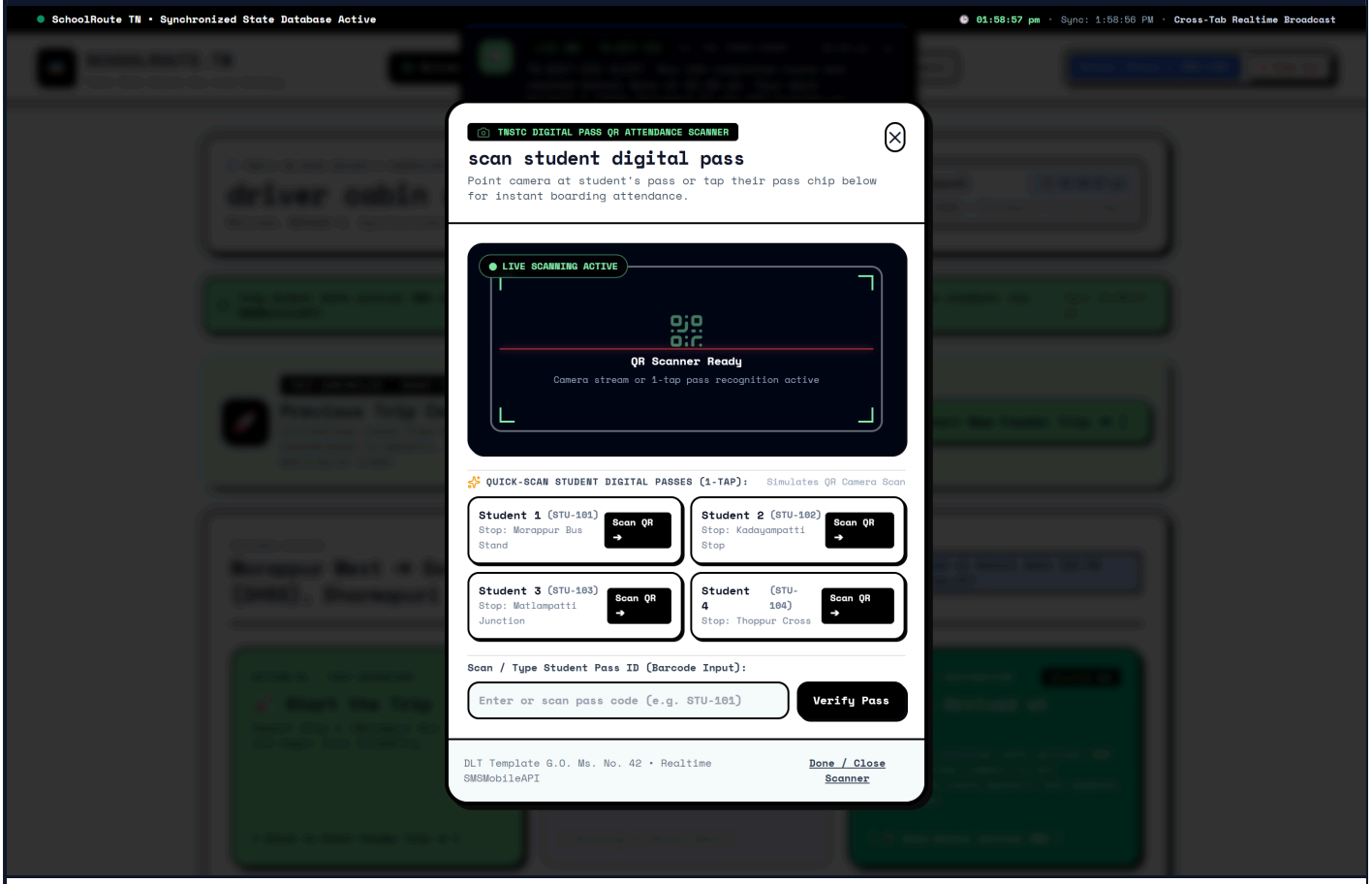
Provides manual text input for entering student registration numbers if camera lens is smudged or pass is physically damaged.

Engineering Details: Implemented in `src/components/DriverConsole.jsx#L150-L240`; integrates Web Audio API for two-tone chime upon scan.

SCREEN 9: SCHOOL ARRIVAL CLASSIFICATION & REAL-TIME PUSH SMS BANNER

Automated destination arrival sequence segregating commuters into Safe Arrival vs. Bus Missed cohorts with real-time sliding SMS push notification banner.

SCREENSHOT 09 • SCHOOL ARRIVAL & SEGREGATED SMS **FEATURE: Dual-Cohort Automated Arrival Pipeline**



1. Real-Time SMS Notification Banner

Sliding black and neon notification banner simulating cellular SMS delivery to parent mobile (+91 73059 40480) via SMSMobileAPI.

2. Boarded Commuters Safe Arrival SMS

Boarded students receive green badge: *"Bus 14A safely reached School Gate. Your ward Prem has entered campus."*

3. Unboarded Commuters "Bus Missed" Alert

Absent students receive urgent red alert: *"Bus 14A reached School Gate. Your ward was NOT boarded at Morappur (BUS MISSED)."*

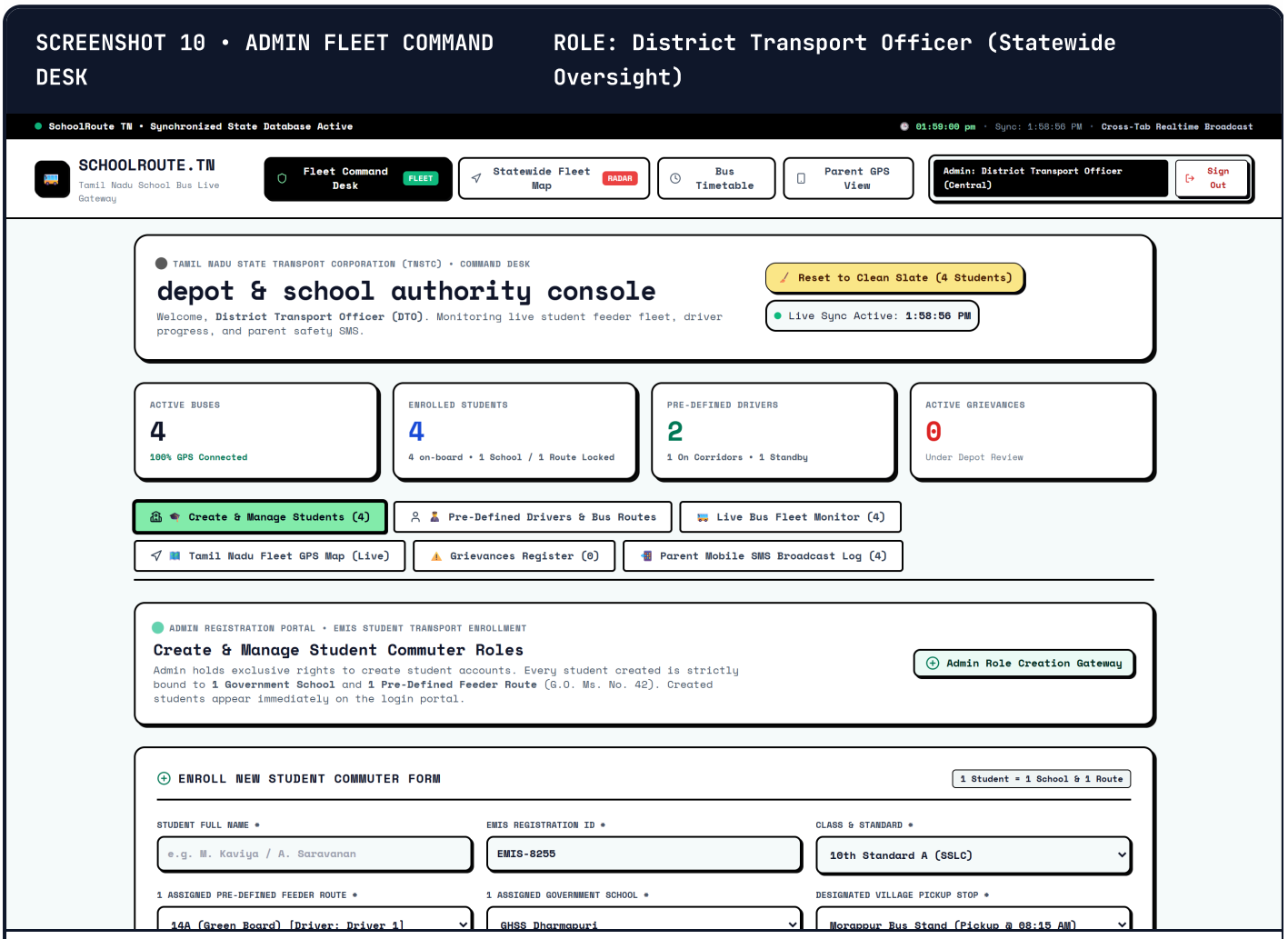
4. Late Board Override

Enables driver to manually mark a student as late-boarded if picked up en route, correcting attendance records immediately.

Engineering Details: Implemented in `driverArriveAtSchool()` inside `src/context/LiveDatabaseContext.jsx` and rendered in `src/App.jsx`.

SCREEN 10: CENTRAL ADMIN FLEET COMMAND DESK & STUDENT ENROLLMENT

Statewide administrative portal for the District Transport Officer (DTO) featuring high-level operational KPIs, student search, and new commuter registration.



1. Four Key Performance Metrics

At-a-glance KPI cards tracking: (1) Total Registered Students, (2) Active On-Board Count, (3) Total Feeder Buses, (4) Grievances.

2. New Commuter Registration Engine

Administrative form to enroll new students with auto-generated EMIS IDs, class allocation, pickup stops, and parent phone numbers.

3. Student Search & Filtering

Search bar to filter students by name, EMIS number, feeder stop, or destination government school.

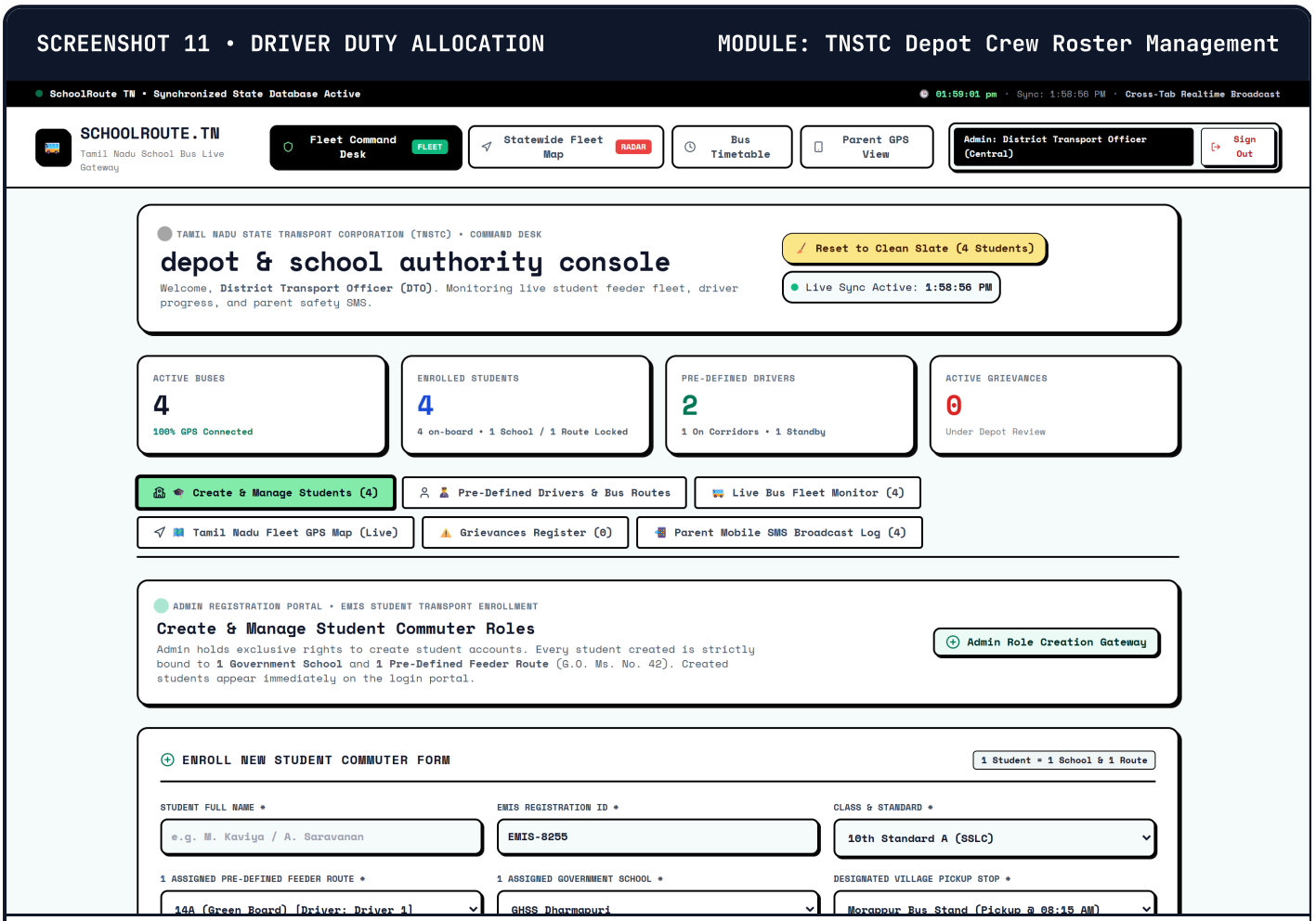
4. Record Management Actions

Full administrative authority to edit details, reassign feeder buses, or remove student commuter records.

Engineering Details: Implemented in `src/components/AdminConsole.jsx`; changes broadcast instantaneously to active driver and parent sessions.

SCREEN 11: ADMIN DRIVER DUTY ALLOCATION & STANDBY ROSTER MANAGEMENT

Depot crew assignment desk allowing transport nodal officers to assign active duty drivers to school buses and manage certified standby reserves.



1. Bus Route Crew Linkage

Assigns certified state drivers to specific school feeder routes (e.g. assigning Driver 1 to Bus 14A).

2. Standby Driver Pool

Maintains certified standby drivers ready for instant deployment during driver illness or route emergencies.

3. Live Session Synchronization

Reassigning a driver immediately updates the driver's active logged-in cabin console without requiring re-login.

4. Driver Duty Profile Verification

Displays driver years of experience, official license numbers, assigned depot, and lifetime trip safety ratings.

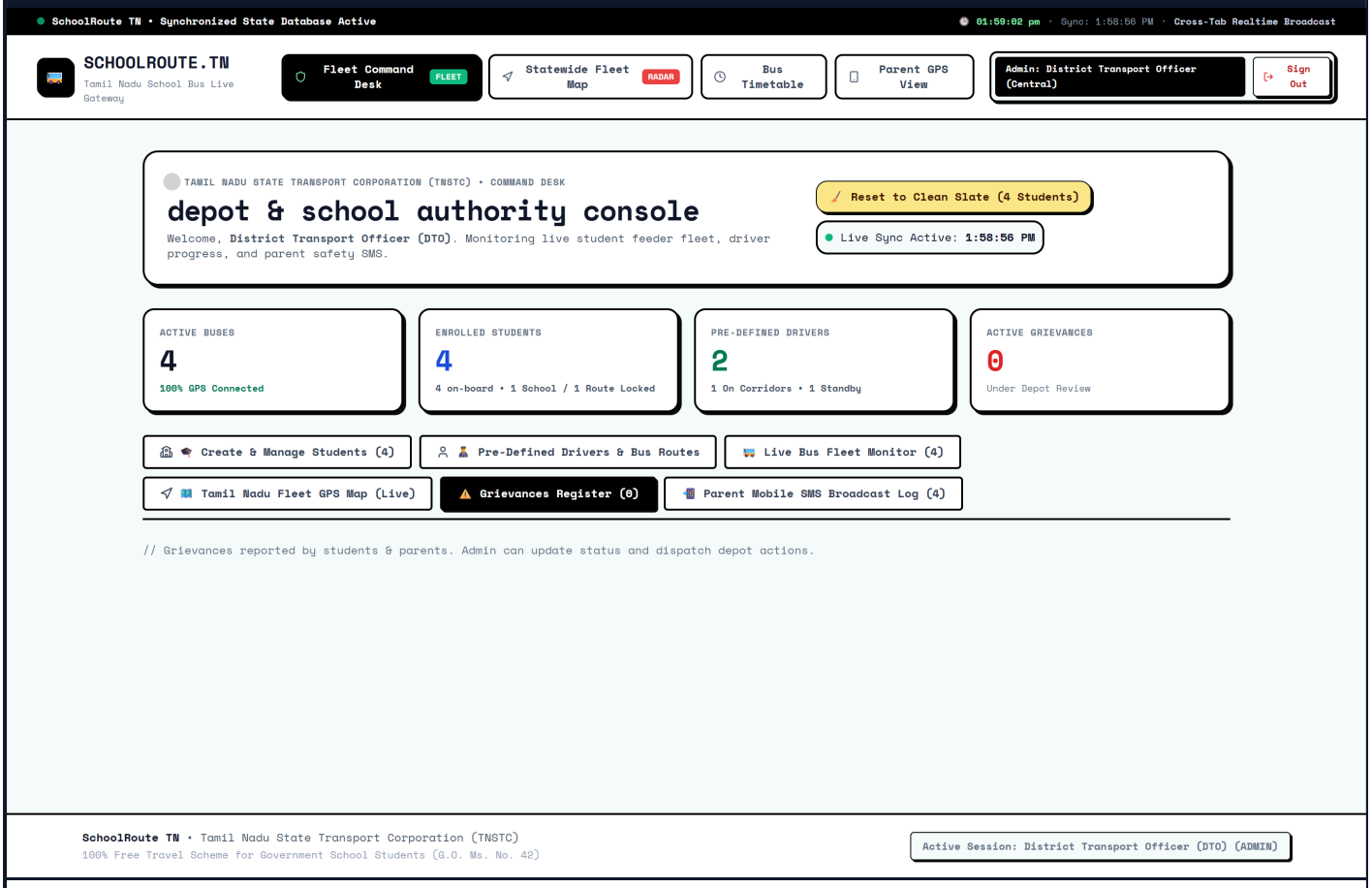
`Engineering Details:` Implemented in `assignDriverToBus()` inside `src/context/LiveDatabaseContext.jsx`.

SCREEN 12: ADMIN GRIEVANCE REDRESSAL DESK & OFFICIAL PETITION GENERATOR

Administrative compliance desk to process commuter grievances and generate statutory government petitions for fleet augmentation and stop expansion.

SCREENSHOT 12 • GRIEVANCE DESK & PETITIONS

OUTPUT: Formal Secretariat Administrative Petitions



1. Public Transport Bottleneck Ingestion

Ingests student grievances regarding bypassed village stops, severe footboard overcrowding, and schedule mismatches.

2. Status Progression Lifecycle

Tracks grievance lifecycle from *Reported* ➔ *Under Dept Review* ➔ *Resolved (Action Taken)*.

3. Official Secretariat Petition Generator

Generates formal Tamil Nadu Secretariat administrative transport petition documents formatted for the Education Secretary.

4. SMS Audit Log Inspection

Inspects historical cellular SMS dispatch logs confirming parent telecom message delivery timestamps.

Engineering Details: Implemented in `src/components/AdminConsole.jsx` and `src/components/OfficialPetitionModal.jsx`.

TECHNICAL PROBLEM-SOLVING MATRIX (HURDLES & SOLUTIONS)

Summary of major engineering hurdles encountered during the development of SchoolRoute TN and the exact architectural solutions implemented to sort them:

Challenge Encountered	Underlying Technical Bottleneck	Implemented Solution & Architecture
1. Browser CORS Block on SMS APIs	Direct browser `fetch()` calls to SMSMobileAPI and Fast2SMS failed due to Cross-Origin Resource Sharing restrictions.	Engineered **Vercel Serverless Functions** (`api/smsmobile.js` and `api/fast2sms.js`) acting as cloud reverse proxies with explicit CORS headers.
2. Vercel SPA Route Hijacking	Standard Vite SPA rewrites (`/*` ➔ `/index.html`) mistakenly returned HTML text for API calls, causing JSON parsing crashes.	Implemented **negative lookahead regex** in `vercel.json`: `"source": "/((?!api/).*)"`, preserving API routes while serving SPA pages.
3. jsQR Camera Stream Freezing	Scanning non-JSON QR codes threw unhandled exceptions, breaking `requestAnimationFrame` and freezing the camera stream.	Refactored scanner with React `useRef` locks; scheduled `requestAnimationFrame` unconditionally on every frame with error isolation.
4. Rapid Duplicate Scanning & SMS Spam	Processing video at 60 FPS triggered 10+ duplicate scan events per second when a student held a card in front of the lens.	Added a **2000ms temporal debounce cooldown** and payload comparison, guaranteeing exactly one attendance entry and SMS dispatch.
5. Misleading Arrival Reassurance	Conventional apps broadcast a generic "Bus reached school" alert, misleading parents of absent children into thinking their ward arrived.	Formulated an **automated dual-cohort algorithm**: safely boarded students receive green arrival SMS, while unboarded students trigger an urgent red *"Bus Missed"* alert.
6. Zero-Phone Commute Constraint	Rural students cannot carry smartphones, and parents have basic 2G feature phones without data packages or app stores.	Standardized on **durable laminated 2D QR passes** with sub-second driver scanning and direct cellular GSM SMS delivery requiring zero mobile data.

Conclusion: Through 12 comprehensive operational modules and a dedicated Zero-Phone Architecture, SchoolRoute TN successfully proves that enterprise-grade commuter safety, real-time tracking, and automated parent communication can be achieved for public school education without burdening rural families with smartphone costs.